

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

Paper 1

Thursday 15 May 2025

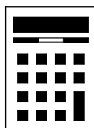
07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



J U N 2 5 9 2 0 2 1 0 1

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9202/1

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ANSWER IN THE SPACES PROVIDED**



Answer **all** questions in the spaces provided.

0 1

This question is about alcohols and carboxylic acids.

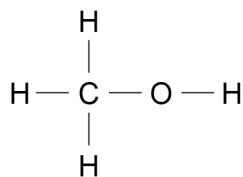
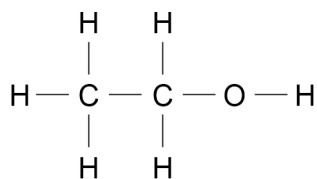
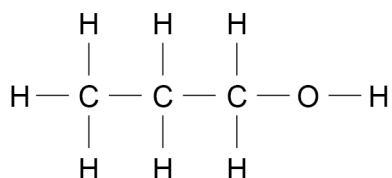
Ethanol is an alcohol.

0 1 . 1

What is the displayed structure of ethanol?

[1 mark]

Tick (✓) **one** box.


☐

☐

☐

0 1 . 2

What is the name of the reaction used to produce ethanol from sugar?

[1 mark]

Tick (✓) **one** box.

Displacement

☐

Fermentation

☐

Neutralisation

☐


0 1 . 3 What is the pH of the solution formed when ethanol is dissolved in water?

[1 mark]

Tick (✓) **one** box.

3

☐

7

☐

10

☐

14

☐

0 1 . 4 Ethanoic acid is a carboxylic acid.

Ethanoic acid can be produced from ethanol.

What type of reaction produces ethanoic acid from ethanol?

[1 mark]

Tick (✓) **one** box.

Combustion

☐

Oxidation

☐

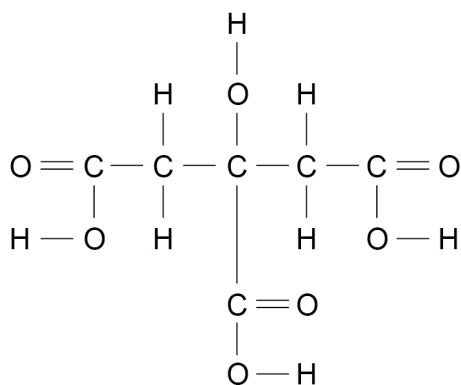
Polymerisation

☐

Citric acid is a carboxylic acid.

Figure 1 shows the displayed structure of citric acid.

Figure 1



0 1 . 5 How many carboxylic acid functional groups are in one molecule of citric acid?

[1 mark]

0 1 . 6 What is the molecular formula of citric acid?

[1 mark]

C ____ H ____ O ____

0 1 . 7 Carboxylic acids are weak acids.

What is meant by a 'weak acid'?

[1 mark]

Question 1 continues on the next page

Turn over ►



0 1 . 8 Carboxylic acids are used in medicines.

Carboxylic acids must be pure when used in medicines.

Suggest why carboxylic acids used in medicines must be pure.

[1 mark]

8

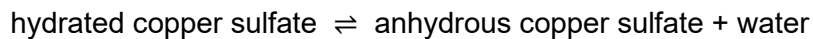


0 2

This question is about copper sulfate.

When hydrated copper sulfate is heated, anhydrous copper sulfate and water are produced.

The equation for the reaction is:

**0 2 . 1**

How does the equation show that the reaction is reversible?

[1 mark]

0 2 . 2

Hydrated copper sulfate is blue.

What colour is anhydrous copper sulfate?

[1 mark]

Tick (✓) **one** box.

Blue

☐

Green

☐

Red

☐

White

☐**0 2 . 3**

How can anhydrous copper sulfate be turned back into hydrated copper sulfate?

[1 mark]

Question 2 continues on the next page

Turn over ►



A student investigated the change in mass when hydrated copper sulfate is heated.

This is the method used.

- 1 Weigh an empty test tube.
- 2 Add 2.00 g of hydrated copper sulfate to the test tube.
- 3 Heat the test tube and contents for 2 minutes.
- 4 Measure the mass of the test tube and contents.
- 5 Determine the mass of the contents of the test tube.
- 6 Repeat steps 1 to 5 three more times.

Table 1 shows the results.

Table 1

	Result 1	Result 2	Result 3	Result 4
Mass of hydrated copper sulfate before heating in grams	2.00	2.00	2.00	2.00
Mass of the contents of the test tube after heating in grams	1.36	1.31	1.38	1.39

0 2 . 4 What is the range of the mass of the contents of the test tube after heating?

Use **Table 1**.

[1 mark]

From _____ to _____

0 2 . 5 Why do the contents of the test tube lose mass?

[1 mark]



0	2	.	6
---	---	---	---

Suggest how the investigation could be improved to make sure the reaction is complete.

[2 marks]

0	2	.	7
---	---	---	---

Copper sulfate solution contains sulfate ions.

Describe a test to show that copper sulfate solution contains sulfate ions.

Give the result of the test.

[3 marks]

Test _____

Result _____

Question 2 continues on the next page

Turn over ►



0	2	.	8
---	---	---	---

3.90 g of hydrated copper sulfate was heated and 2.50 g of anhydrous copper sulfate was produced.

A different mass of hydrated copper sulfate was heated and 15.12 g of water was produced.

Calculate the mass of hydrated copper sulfate needed to produce 15.12 g of water.

The equation for the reaction is:

hydrated copper sulfate $\rightleftharpoons$ anhydrous copper sulfate + water

[3 marks]

Mass = _____ g

13



0 3

This question is about nanoparticles.

0 3 . 1

What size are nanoparticles?

[1 mark]Tick (✓) **one** box.

1–100 nm

☐

1–1000 nm

☐

1–10 000 nm

☐**0 3 . 2**

How many atoms are in a nanoparticle?

[1 mark]Tick (✓) **one** box.

A few hundred

☐

A few thousand

☐

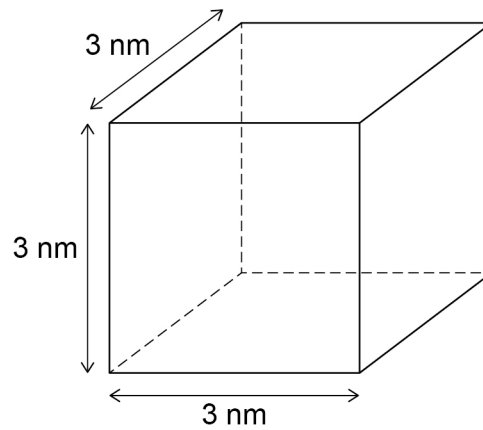
A few million

☐**Question 3 continues on the next page****Turn over ►**

0 3 . 3

Figure 2 shows a cubic nanoparticle.

Figure 2



The surface area of the cubic nanoparticle is 54 nm^2 .

The volume of a cube is given by the equation:

$$\text{volume} = \text{width} \times \text{length} \times \text{height}$$

Calculate:

- the volume of the cubic nanoparticle
- the simplest surface area : volume ratio of the cubic nanoparticle.

[4 marks]

Volume = _____ nm^3

Simplest surface area : volume ratio = _____ : 1



0	3	.	4
---	---	---	---

Copper oxide nanoparticles are used in hospitals to kill bacteria on surfaces.

Copper oxide nanoparticles are used instead of larger particles of copper oxide.

Suggest **one** reason why.

[1 mark]

7

Turn over for the next question

Turn over ►



0 4

The current periodic table of elements is based on the work of Mendeleev.

0 4 . 1

How many different elements are there in the current periodic table?

[1 mark]Tick (✓) **one** box.

About 50

☐

About 100

☐

About 500

☐

About 1000

☐**0 4 . 2**

Complete the sentence.

Choose the answer from the box.

atomic number**mass number****neutron number****[1 mark]**

The current periodic table is arranged in order of _____.



0 4 . 3

Figure 3 shows Group 1 of the periodic table Mendeleev produced in 1869.**Figure 3**

Group 1	
H	
Li	
Na	
K	Cu
Rb	Ag

Give **three** differences between Group 1 in Mendeleev's periodic table and Group 1 in the current periodic table.

Use the periodic table.

[3 marks]

- 1 _____

- 2 _____

- 3 _____

Question 4 continues on the next page

Turn over ►



0 4 . 4 Mendeleev left gaps in his periodic table for elements that had **not** been discovered.

He predicted some properties of these elements.

Table 2 shows the properties Mendeleev predicted for element **X**.

Table 2

	Element X
Melting point in °C	Low
Density in g/cm ³	6.0
Formula of oxide	X ₂ O ₃

Table 3 shows some of the properties of elements that were discovered after Mendeleev published his periodic table.

Table 3

	Scandium (Sc)	Gallium (Ga)	Germanium (Ge)
Melting point in °C	1541	30	938
Density in g/cm ³	3.0	5.9	5.3
Formula of oxide	Sc ₂ O ₃	Ga ₂ O ₃	GeO ₂

Which element matches Mendeleev's predictions for element **X**?

Give **three** reasons for your answer.

Use **Table 2** and **Table 3**.

[4 marks]

Element _____

Reason 1 _____

Reason 2 _____

Reason 3 _____



0 4 . 5

The existence of element **D** was also predicted by Mendeleev.

A compound containing element **D** has the formula D_2O_7

D is **not** the symbol of the element.

The relative formula mass of the compound is 484

Determine the identity of element **D**.

Use the periodic table.

Relative atomic mass (A_r): O = 16

[3 marks]

Element **D** = _____

12

Turn over for the next question

Turn over ►



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0 5

Diamond and graphite are made of carbon atoms.

0 5 . 1

Which is the maximum number of bonds that one carbon atom can form?

[1 mark]

Tick (✓) **one** box.

2

☐

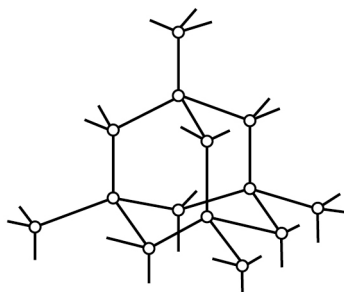
3

☐

4

☐

5

☐**Figure 4** represents part of the structure of a diamond.**Figure 4**

0 5 . 2

What is a property of diamond?

[1 mark]

Tick (✓) **one** box.

Conducts electricity

☐

Dissolves in water

☐

Very hard

☐

Turn over ►



0 5 . 3 What is the approximate melting point of diamond?

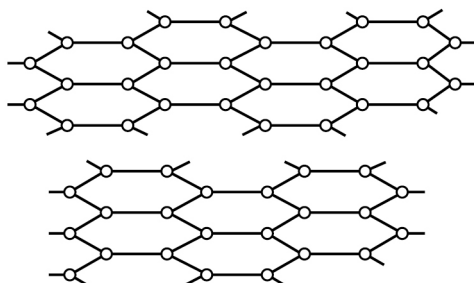
[1 mark]

Tick (✓) **one** box.

−4000 °C ☐ −40 °C ☐ 40 °C ☐ 4000 °C ☐

0 5 . 4 **Figure 5** represents part of the structure of graphite.

Figure 5



Explain why graphite is:

- soft and slippery
- a good conductor of electricity.

Answer in terms of structure and bonding.

[6 marks]



Extra space

9

Turn over for the next question

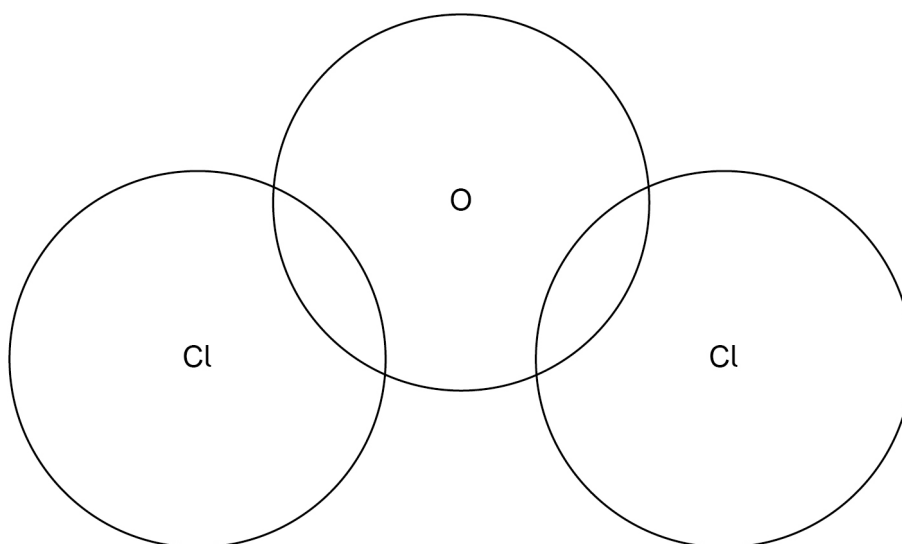
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0 6

Elements react to form compounds.

0 6 . 1

Chlorine atoms react with oxygen atoms to form the covalent compound Cl_2O Complete **Figure 6** to show the arrangement of the outer shell electrons in Cl_2O Use the symbols $\bullet$ or $\times$ to represent the electrons.**[2 marks]****Figure 6**

Magnesium atoms react with chlorine atoms to produce the ionic compound magnesium chloride.

0 6 . 2

Suggest **one** reason why the reaction between magnesium and chlorine needs to be carried out in the absence of air.**[1 mark]**



0	6	.	3
---	---	---	---

Describe what happens when magnesium atoms react with chlorine atoms to produce magnesium chloride.

[4 marks]

Extra space _____

0	6	.	4
---	---	---	---

Magnesium chloride contains magnesium ions.

Describe the test for magnesium ions.

Give the result of the test.

[2 marks]

Test _____

Result _____

Question 6 continues on the next page

Turn over ►



0 6 . 5 Magnesium reacts with element **Q** to form an ionic compound (MgQ).

Element **Q** forms ions with a 2^- charge.

The ions of element **Q** have the same electronic arrangement as a krypton atom.

Identify element **Q**.

Use the periodic table.

[1 mark]

0 6 . 6 A different magnesium compound contains:

- 8.40 g magnesium
- 16.80 g oxygen
- 22.40 g sulfur.

Determine the empirical formula of the compound.

Relative atomic masses (A_r): Mg = 24 O = 16 S = 32

[4 marks]

Empirical formula = _____

14



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Turn over ►



2 5

0 7

Ammonia (NH_3) is made using the Haber process.

Nitrogen and hydrogen react to produce ammonia.

0 7 . 1

What is the source of the nitrogen and of the hydrogen used in the Haber process?

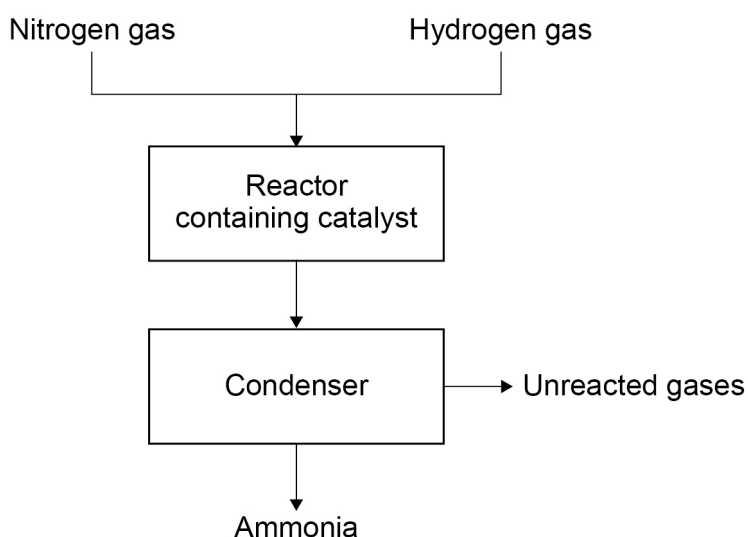
[2 marks]

Nitrogen _____

Hydrogen _____

Figure 7 shows a flow diagram for the production of ammonia in the Haber process.

Figure 7

**0 7 . 2**

Name the catalyst used in the Haber process.

[1 mark]



0 7 . 3 The mixture of gases from the reactor is separated in the condenser.

Table 4 shows the boiling points of the substances shown in **Figure 7**.

Table 4

Substance	Boiling point in °C
Nitrogen	−196
Hydrogen	−253
Ammonia	−33

Suggest the temperature used in the condenser.

Give **one** reason for your answer.

[2 marks]

Temperature _____

Reason _____

0 7 . 4 What happens to the mixture of unreacted gases when the gases leave the condenser?

Use **Figure 7**.

[1 mark]

Question 7 continues on the next page

Turn over ►



The reaction to produce ammonia is a reversible reaction in a closed system.

0 7 . 5 Suggest what is meant by the term 'closed system'.

[1 mark]

0 7 . 6 Give **one** feature of a reversible reaction when equilibrium is reached.

[1 mark]

The equation for the reaction to produce ammonia is:

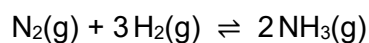


Table 5 shows the percentage (%) of ammonia present at equilibrium using different temperatures and different pressures.

Table 5

Pressure in atmospheres	Percentage (%) of ammonia present at equilibrium				
	Temperature in °C				
	100 °C	200 °C	300 °C	400 °C	500 °C
25	92	64	27	9	3
50	95	74	40	15	6
100	97	82	53	25	11
200	98	89	67	39	30
400	99	95	80	55	32



0	7	.	7
---	---	---	---

Explain the effect of increasing the temperature on the percentage of ammonia at equilibrium.

Use **Table 5**.

[3 marks]

0	7	.	8
---	---	---	---

Explain the effect of increasing the pressure on the percentage of ammonia at equilibrium.

Use **Table 5**.

[3 marks]

14

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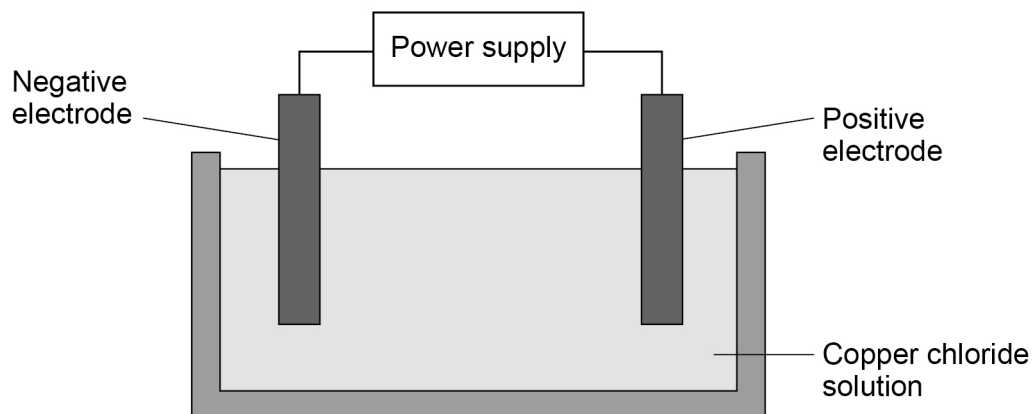


0 8

This question is about electrolysis.

Figure 8 shows the apparatus used for the electrolysis of copper chloride solution.

Figure 8



Copper forms on the negative electrode.

0 8 . 1

Why does copper chloride solution conduct electricity?

[1 mark]

0 8 . 2

The apparatus in **Figure 8** can be used to electroplate metal objects with copper.

Give **one** reason why some metal objects are electroplated with copper.

[1 mark]

0 8 . 3

Chlorine gas (Cl_2) is produced at the positive electrode.

Complete the half equation for the production of chlorine gas at the positive electrode.

[2 marks]



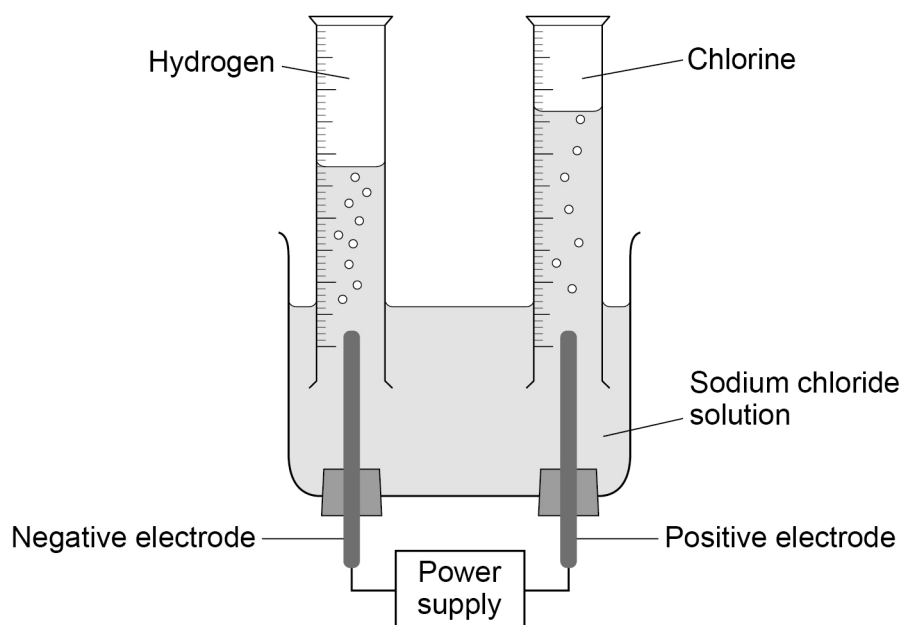
Turn over ►



A student investigated the electrolysis of sodium chloride solution.

Figure 9 shows the apparatus.

Figure 9



The electrolysis of sodium chloride solution produces hydrogen and chlorine.

Sodium hydroxide solution is also produced.

0 8 . 4 What is the pH of sodium hydroxide solution?

[1 mark]

0 8 . 5 Sodium hydroxide solution is used as a reagent for the chemical industry.

Name **one** substance produced using sodium hydroxide solution.

[1 mark]



08.6

Calculate the mass of sodium hydroxide (NaOH) in 600 cm³ of 0.350 mol/dm³ sodium hydroxide solution.

Relative formula mass (M_r): NaOH = 40

[4 marks]

Mass = _____ g

08.7

Why is hydrogen gas produced at the negative electrode instead of sodium metal in the electrolysis of sodium chloride solution?

[1 mark]

08.8

Explain how hydrogen gas is produced at the negative electrode in the electrolysis of sodium chloride solution.

[2 marks]

13

END OF QUESTIONS



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